· Upper-Confidence-bound Action Selection:

At t^{th} time step action a has been chosen $N_a(t)$ times prior to t, yeilding rewards $R_1, R_2, \ldots, R_{N_t(a)}$, then its value is estimated to be:

$$Q_t(a) = \frac{R_1 + R_2 + \ldots + R_{N_t(a)}}{N_t(a)}$$

Greedy action:

$$A_t = \arg\max_a Q_t(a)$$

However, estimates of the action values are uncertain. It would be better to select among the non-greedy actions according to their potential for actually being optimal, taking into account both how close their estimates are to being maximal and the uncertainties in those estimates. One effective way of doing this.

uncertainty

$$A_t = \arg\max_a \left[Q_t(a) + \boxed{c\sqrt{\frac{\ln t}{N_t(a)}}} \right]$$

t: time; If $N_t(a) = 0$, then a is considered to be a maximizing action.

c is the confidence level.

Assignment :

Repeat 10-armed testbed with

$c = 1, 2, 3, 4, 5 \ldots\ldots$

$\epsilon = 0.1, 0.001 \ldots$